

RICOVIR

Tenofovir Disoproxil Fumarate 300 mg Tablets

1. NAME OF THE MEDICINAL PRODUCT

Tenofovir Disoproxil Fumarate Film Coated Tablet 300 mg

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains:

Tenofovir disoproxil fumarate 300 mg

3. PHARMACEUTICAL FORM

Film-coated tablets.

Light blue colored, round biconvex film-coated tablet debossed with “153” on one side and “M” on other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

For HIV-1 Infection

Tenofovir Disoproxil Fumarate 300 mg Tablets is indicated in combination with other antiretroviral agents for the treatment of HIV-1 infection in adults and paediatric patients 12 years of age and older.

The following points should be considered when initiating therapy with Tenofovir Disoproxil Fumarate 300 mg Tablets for the treatment of HIV-1 infection:

- Tenofovir Disoproxil Fumarate 300 mg Tablets should not be used in combination with preparations containing emtricitabine/tenofovir disoproxil fumarate or efavirenz/emtricitabine/tenofovir disoproxil fumarate.

For Chronic Hepatitis B

Tenofovir Disoproxil Fumarate 300 mg Tablets is indicated for the treatment of chronic hepatitis B in adults.

- The following points should be considered when initiating therapy with Tenofovir Disoproxil Fumarate 300 mg Tablets for the treatment of HBV infection:
- The indication is based primary on data from treatment of subjects who were nucleoside-treatment-naïve and a smaller number of subjects who had previously received lamivudine or adefovir dipivoxil. Subjects were adults with HBeAg-positive and HBeAg-negative chronic hepatitis B with compensated liver disease.
- Tenofovir Disoproxil Fumarate 300 mg Tablets was evaluated in a limited number of subjects with chronic hepatitis B and decompensated liver disease.
- The numbers of subjects in clinical trials who had lamivudine or adefovir associated substitutions at baseline were too small to reach conclusions of efficacy.

4.2 Posology and method of administration

Recommended Dose in Adults

For the treatment of HIV-1 or chronic hepatitis B : The dose is one 300 mg Tenofovir Disoproxil Fumarate 300 mg Tablets once daily taken orally, without regard to food. In the treatment of chronic hepatitis B, the optimal duration of treatment is unknown. Dose Adjustment for Renal Impairment in Adults

Significantly increased drug exposures occurred when Tenofovir Disoproxil Fumarate 300 mg Tablets was administered to subjects with moderate to severe renal impairments. Therefore, the dosing interval of Tenofovir Disoproxil Fumarate 300 mg Tablets should be adjusted in patients with baseline creatinine clearance <50 mL/min using the recommendations in Table 1. These dosing interval recommendations are based on modeling of single-dose pharmacokinetic data in non-HIV and NON-HBV infected subjects with varying degrees of renal impairment, including end-stage renal disease requiring hemodialysis. The safety and effectiveness of these dosing interval adjustment recommendations have not been clinically evaluated in patients with moderate or severe renal impairment, therefore clinical response to treatment and renal function should be closely monitored in these patients. No dose adjustment is necessary for patients with mild renal impairment (creatinine clearance 50-80 mL/min. Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients with mild renal impairment.

Table 1: Dosage Adjustment for Patients with Altered Creatinine Clearance

	Creatinine Clearance (mL/min) ^a			Hemodialysis Patients
	≥50	30-49	10-29	
Recommended 300 mg Dosing Interval	Every 24 hours	Every 48 hours	Every 72 to 96 hours	Every 7 days or after a total of approximately 12 hours of dialysis ^b

^a Calculated using ideal (lean) body weight

^b Generally once weekly assuming three hemodialysis session a week of approximately 4 hours duration.

Tenofovir Disoproxil Fumarate 300 mg Tablets should be administered following completion of dialysis. The pharmacokinetics of tenofovir have not been evaluated in non-hemodialysis patients with creatinine clearance <10 mL/min; therefore, no dosing recommendation is available for these patients. No data are available to make dose recommendations in pediatric patients 12 years of age and older with renal impairment.

4.3 Contraindications

Known hypersensitivity to tenofovir, tenofovir disoproxil fumarate, or to any of the excipients in the film-coated tablets. Tenofovir Disoproxil Fumarate 300 mg Tablets must not be administered to children less than 12 years of age until further data become available. Tenofovir Disoproxil Fumarate 300 mg Tablets should not be administered concurrently with fixed dose combination tablets containing tenofovir disoproxil fumarate or adefovir dipivoxil.

4.4 Special warnings and special precautions for use

General: Tenofovir Disoproxil Fumarate has not been studied in patients under the age of 12 years or in patients over the age of 65 years. Elderly patients are more likely to have decreased renal function, therefore caution should be exercised when treating elderly patients with tenofovir Disoproxil Fumarate (see below). HIV antibody testing should be offered to all HBV infected patients before initiating tenofovir therapy (see below *Co-infection with HIV-1 and hepatitis B*). In turn, HBV antibody testing should be offered to all HIV infected patients before initiating tenofovir therapy. In the treatment of chronic hepatitis B, limited data are currently available in immuno-suppressed patients or those receiving immuno-suppressive regimens, orthotrophic liver transplant patients and patients coinfectd with the hepatitis C or D virus. As clinical studies have not included sufficient numbers of subjects to determine whether these patients respond differently to Tenofovir Disoproxil Fumarate 300 mg Tablets chronic hepatitis B therapy, such patients should be closely monitored.

Use in children:

The safety and efficacy of Tenofovir Disoproxil Fumarate 300 mg Tablets has not been established in children less than 12 years of age. The clinical relevance of the long term effects of tenofovir disoproxil fumarate treatment on BMD are unknown, and at present the data on the reversibility of renal toxicity effects is limited. Therefore, a multidisciplinary approach is recommended to consider the benefit/risk balance of treatment.

As hepatitis B is a chronic disease of the liver, ongoing clinical monitoring is recommended. *Use in the elderly:*

In general, dose selection for the elderly patient should be cautious, keeping in mind the greater frequency of decreased hepatic, renal or cardiac function, and of concomitant disease or other drug therapy.

Impaired renal function:

Dosing interval adjustment is required in all patients with creatinine clearance <50 mL/min (See DOSAGE AND ADMINISTRATION). The proposed dose interval modifications are based on limited data and may not be optimal. Therefore, clinical response to treatment and renal function should be closely monitored in these patients.

Renal impairment, including cases of acute renal failure and Fanconi syndrome (renal tubular injury with severe hypophosphataemia), has been reported in association with the use of Tenofovir Disoproxil Fumarate 300 mg Tablets.

Tenofovir Disoproxil Fumarate 300 mg Tablets should be avoided with concurrent or recent use of a nephrotoxic agent.

It is recommended that creatinine clearance is calculated in all patients prior to initiating therapy and, as clinically appropriate, during Tenofovir Disoproxil Fumarate 300 mg Tablets therapy. Patients at risk for, or with a history of, renal dysfunction, including patients who have previously experienced renal events while receiving adefovir dipivoxil, should be routinely monitored for changes in serum creatinine and phosphorus.

Lactic Acidosis/Severe Hepatomegaly with Steatosis:

Lactic acidosis and severe hepatomegaly with steatosis, including fatal cases, have been reported with the use of antiretroviral nucleoside analogues alone or in combination, including tenofovir disoproxil fumarate, in the treatment of HIV infection. A majority of these cases have been reported in women. The preclinical and clinical data suggest that the risk of occurrence of lactic acidosis, a class effect of nucleoside analogues is low for tenofovir disoproxil fumarate. However, as tenofovir is structurally related to nucleoside analogues, this risk cannot be excluded. Caution should be exercised when administering Tenofovir Disoproxil Fumarate 300 mg Tablets to any patient, and particularly to those with known risk factors for liver disease. Treatment with Tenofovir Disoproxil Fumarate 300 mg Tablets should be suspended in any patient who develops clinical or laboratory findings suggestive of lactic acidosis or hepatotoxicity.

Decreases in Bone Mineral Density

In reference to a published trial report , decreases in bone mineral density at the lumbar spine, increases in levels of 4 biochemical markers of bone metabolism, and increased serum parathyroid hormone levels were reported in HIV-infected patients receiving tenofovir concomitantly with lamivudine and efavirenz; these effects also were reported to a lesser extent in patients who received a regimen of stavudine, lamivudine, and efavirenz. With the exception of bone-specific alkaline phosphatase concentrations, these changes generally remained within the normal range and the clinical importance is unknown. There were 4 bone fractures reported in patients receiving the regimen that contained tenofovir and 6 reported in those receiving the regimen that did not include tenofovir. Bone monitoring should be considered for HIV-infected patients who have a history of pathologic bone fracture or are at substantial risk for osteopenia.Although the effect of supplementation with calcium and vitamin D was not studied, such supplementation may be beneficial for all patients.If bone abnormalities are suspected, appropriate consultation should be obtained.Osteomalacia and decreases in bone mineral density have been reported in toxicology studies in juvenile animals given high doses of tenofovir.

4.5 Interaction with other medicinal products and other forms of interaction

Interaction studies have only been performed in adults.

Based on the results of *in vitro* experiments and the known elimination pathway of tenofovir, the potential for CYP450 mediated interactions involving tenofovir with other medicinal products is low.

Tenofovir is excreted renally, both by filtration and active secretion via the anionic transporter (hOAT1). Co-administration of tenofovir disoproxil fumarate with other medicinal products that are also actively secreted via the anionic transporter (e.g. didofovir) may result in increased concentrations of tenofovir or of the co-administered medicinal product. *Concomitant antiretroviral medicinal products*

Emtricitabine, lamivudine, indinavir, efavirenz, nelfinavir, and saquinavir (ritonavir boosted): co-administration with tenofovir disoproxil fumarate did not result in any clinically relevant interaction. When tenofovir disoproxil fumarate was administered with lopinavir/ritonavir, no changes were observed in the pharmacokinetics of lopinavir and ritonavir. Tenofovir AUC was increased by approximately 30% when tenofovir disoproxil fumarate was administered with lopinavir/ritonavir. Higher tenofovir concentrations could potentiate tenofovir associated adverse events, including renal disorders.

When didanosine gastro-resistant capsules were administered 2 hours prior to or concurrently with tenofovir disoproxil fumarate, the AUC for didanosine was on average increased by 48% and 60% respectively. The mean increase in the AUC of didanosine was 44% when the buffered tablets were administered 1 hour prior to tenofovir. In both cases the pharmacokinetic parameters for tenofovir administered with a light meal were unchanged. The co-administration of tenofovir disoproxil fumarate and didanosine is not recommended.

When tenofovir disoproxil fumarate was administered with atazanavir, a decrease in concentrations of atazanavir was observed (decrease of 25% and 40% of AUC and Cmin respectively compared to atazanavir 400 mg). When ritonavir was added to atazanavir, the negative impact of tenofovir on atazanavir Cmin was significantly reduced, whereas the decrease of AUC was of the same magnitude (decrease of 25% and 26% of AUC and Cmin respectively compared to atazanavir/ritonavir 300/100 mg). The co-administration of atazanavir/ritonavir with tenofovir resulted in increased exposure to tenofovir. Higher tenofovir concentrations could potentiate tenofovir-associated adverse events, including renal disorders. The co-administration of atazanavir with ritonavir in combination with tenofovir has been substantiated in a clinical study.

Other interactions

Co-administration of tenofovir disoproxil fumarate, methadone, ribavirin, adefovir dipivoxil or the hormonal contraceptive norgestimate/ethinyl oestradiol did not result in any pharmacokinetic interaction.

Tenofovir disoproxil fumarate must be taken with food, as food enhances the bioavailability of tenofovir.

4.6 Pregnancy and lactation

Pregnancy

No clinical data on exposed pregnancies are available for tenofovir disoproxil fumarate. Animal studies do not indicate direct or indirect harmful effects of tenofovir disoproxil fumarate with respect to pregnancy, foetal development, parturition or postnatal development. Tenofovir disoproxil fumarate should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus. However, given that the potential risks to developing human foetuses are unknown, the use of tenofovir disoproxil fumarate in women of childbearing potential must be accompanied by the use of effective contraception. *Lactation* In animal studies it has been shown that tenofovir is excreted into milk. It is not known whether tenofovir is excreted in human milk. Therefore, it is recommended that mothers being treated with tenofovir disoproxil fumarate do not breast-feed their infants.

As a general rule, it is recommended that HIV infected women do not breast-feed their infants in order to avoid transmission of HIV to the infant.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. However, patients should be informed that dizziness has been reported during treatment with tenofovir disoproxil fumarate.

4.8 Undesirable effects

Approximately one third of patients can be expected to experience adverse reactions following treatment with tenofovir disoproxil fumarate in combination with other antiretroviral agents.

These reactions are usually mild to moderate gastrointestinal events.

Metabolism and nutrition disorders:

Very common: hypophosphataemia

Rare: lactic acidosis

Nervous system disorders:

Very common: dizziness

Bone effects

Common: decrease in bone mineral density

Respiratory, thoracic and mediastinal disorders:

Very rare: dyspnoea

Gastrointestinal disorders:

Very common: diarrhoea, nausea, vomiting

Common: flatulence

Rare: pancreatitis

Hepatobiliary disorders:

Rare: increased transaminases

Very rare: hepatitis

Musculoskeletal and connective tissue disorders:

Not known: myopathy, osteomalacia (both associated with proximal renal tubulopathy) *Renal and urinary disorders:*

Rare: renal failure, acute renal failure, proximal tubulopathy (including Fanconi syndrome), increased creatinine

Very rare: acute tubular necrosis

In addition, there have been post-marketing reports of nephritis and nephrogenic diabetes insipidus.

General disorders and administration site conditions:

Very rare: asthenia

Approximately 1% of tenofovir disoproxil fumarate treated patients discontinued treatment due to the gastrointestinal events. Combination antiretroviral therapy has been associated with metabolic abnormalities such as hypertriglyceridaemia, hypercholesterolaemia, insulin resistance, hyperglycaemia and hyperlactataemia.

Combination antiretroviral therapy has been associated with redistribution of body fat (lipodystrophy) in HIV patients including the loss of peripheral and facial subcutaneous fat, increased intra-abdominal and visceral fat, breast hypertrophy and dorsocervical fat accumulation (buffalo hump).

In HIV infected patients with severe immune deficiency at the time of initiation of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic infections may arise.

Cases of osteonecrosis have been reported, particularly in patients with generally acknowledged risk factors, advanced HIV disease or long-term exposure to combination antiretroviral therapy (CART). The frequency of this is unknown.

4.9 Overdose

If overdose occurs the patient must be monitored for evidence of toxicity (see sections 4.8 and 5.3), and standard supportive treatment applied as necessary.

Tenofovir can be removed by haemodialysis; the median haemodialysis clearance of tenofovir is 134 mL/min. The elimination of tenofovir by peritoneal dialysis has not been studied.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Nucleoside and nucleotide reverse transcriptase inhibitors, ATC code: J05AF07

Mechanism of action: Tenofovir disoproxil fumarate is the fumarate salt of the prodrug tenofovir disoproxil. Tenofovir disoproxil is absorbed and converted to the active substance tenofovir, which is a nucleoside monophosphate (nucleotide) analogue. Tenofovir is then converted to the active metabolite, tenofovir diphosphate, by constitutively expressed cellular enzymes through two phosphorylation reactions in both resting and activated T cells.

 Mylan Laboratories Limited Packaging Development	Artwork Implementation Schedule Check (✓) whichever is applicable (Approval is not valid without following details)					Date of Issue				Issued By				
						Date of Return								
					Material Code		75091542	Supersedes	75065232	Market	Malaysia			
					Description		LIT. RICOVIR TABS 300 mg (36 Months) MY V2							
					Component		Printed Literature		Actual Size	Flat- 272 x 312 mm; Folded- 34 x 39 mm				
					Substrate		60gsm Century Maplitho Paper							
					Design & Style		Supply in Folded form as Proposed Size (with tape)							
					Reason for Issue		Change in Mylan Healthcare address							
					Printing Pantone Nos		1	BLACK	2	NA	3	NA	4	NA
							5	NA	6	NA	7	NA	8	NA
					Non Printing			Die Line		NA		NA		NA
Proof No.		1	2	3	4	5	Prepared By		Checked By			Approved By		
Remarks		Revised A/w	x	x	x	x	Packaging Development		Packaging Development		Production		Regulatory Affairs	
													Quality Assurance	
SOP-000565164-FORM-000565208-A01-03-01-20					Final Date		dd/mm/yy							

Tenofovir diphosphate has an intracellular half-life of 10 hours in activated and 50 hours in resting peripheral blood mononuclear cells (PBMCs). Tenofovir diphosphate inhibits viral polymerases by direct binding competition with the natural deoxyribonucleotide substrate and, after incorporation into DNA, by DNA chain termination. Tenofovir diphosphate is a weak inhibitor of cellular polymerases α , β , and γ , with kinetic inhibition constants (Ki) that are > 200-fold higher against human DNA polymerase α (5.2 $\mu\text{mol/l}$) and > 3,000-fold higher against human DNA polymerase β and γ (81.7 and 59.5 $\mu\text{mol/l}$, respectively) than its Ki against HIV-1 reverse transcriptase (0.02 $\mu\text{mol/l}$). At concentrations of up to 300 $\mu\text{mol/l}$, tenofovir has also shown no effect on the synthesis of mitochondrial DNA or the production of lactic acid in *in vitro* assays.

Pharmacodynamic effects: Tenofovir has *in vitro* antiviral activity against retroviruses and hepadnaviruses.

The concentration of tenofovir required for 50% (IC₅₀) of the wild-type laboratory strain HIV-1IIB is 1-6 $\mu\text{mol/l}$ in lymphoid cell lines and 1.1 $\mu\text{mol/l}$ against primary HIV-1 subtype B isolates in PBMCs. Tenofovir is also active against HIV-1 subtypes A, C, D, E, F, G, and O and against HIVBaL in primary monocyte/macrophage cells. Tenofovir shows activity *in vitro* against HIV-2, with an IC₅₀ of 4.9 $\mu\text{mol/l}$ in MT-4 cells and against hepatitis B virus, with an IC₅₀ of 1.1 $\mu\text{mol/l}$ in HepG2 2.2.15 cells.

The activity of tenofovir remains within two-fold of wild-type IC₅₀ against recombinant HIV-1 expressing didanosine resistance (L74V), zalcitabine resistance (T69D), and multinucleoside drug resistance (Q151M complex) mutations. The activity of tenofovir against HIV-1 strains with zidovudine-associated mutations appears to depend on the type and number of these resistance mutations. In the presence of mutation T215Y, a two-fold increase of the IC₅₀ was observed. In 10 samples which had multiple zidovudine-associated mutations (mean 3.4), a mean 3.7-fold increase of the IC₅₀ was observed (range 0.8 to8.4). Multinucleoside resistant HIV-1 with T69S double insertions have reduced susceptibility to tenofovir (IC₅₀> 10-fold). Tenofovir shows full activity against non-nucleoside reverse transcriptase inhibitor resistant HIV-1 with K103N or Y181C mutations. Cross-resistance to protease inhibitor resistance mutations is not expected due to the different viral enzymes targeted.

Strains of HIV-1 with 3- to 4-fold reduced susceptibility to tenofovir and a K65R mutation in reverse transcriptase have been selected *in vitro*. The K65R mutation in reverse transcriptase can also be selected by zalcitabine, didanosine, and abacavir, and causes reduced susceptibility to zalcitabine, didanosine, abacavir, and lamivudine (14-, 4-, 3-, and 25-fold, respectively). Tenofovir disoproxil fumarate should be avoided in antiretroviral experienced patients with strains harbouring the K65R mutation.

The clinical activity of tenofovir disoproxil fumarate has not been determined against hepatitis B virus (HBV) in humans. It is unknown whether treatment of patients co-infected with HIV-1 and HBV will result in the development of HBV resistance to tenofovir disoproxil fumarate or other medicinal products.

Clinical efficacy: The effects of tenofovir disoproxil fumarate in treatment-experienced and treatment-naïve HIV-1 infected adults have been demonstrated in trials of 48 weeks duration in treatment-experienced HIV-1 infected adults.

In study GS-99-907, 550 treatment-experienced patients were treated with placebo or tenofovir disoproxil 245 mg (as fumarate) for 24 weeks. The mean baseline CD4 cell count was 427 cells/mm3, the mean baseline plasma HIV-1 RNA was 3.4 log₁₀ copies/ml (78% of patients had a viral load of < 5,000 copies/ml) and the mean duration of prior HIV treatment was 5.4 years. Baseline genotypic analysis of HIV isolates from 253 patients revealed that 94% of patients had HIV-1 resistance mutations associated with nucleoside reverse transcriptase inhibitors, 58% had mutations associated with protease inhibitors and 48% had mutations associated with nonnucleoside reverse transcriptase inhibitors.

At week 24 the time-weighted average change from baseline in log₁₀ plasma HIV-1 RNA levels (DAVG24) was -0.03 log₁₀ copies/ml and -0.61 log₁₀ copies/ml for the placebo and tenofovir disoproxil 245 mg (as fumarate) recipients (p < 0.0001). Patients whose HIV expressed 3 or more thymidine-analogue associated mutations (TAMs) that included either the M41L or L210W reverse transcriptase mutation showed reduced susceptibility to tenofovir disoproxil 245 mg (as fumarate) therapy. The virological response was substantially decreased in patients with viral strains of > 10-fold zidovudine phenotypic resistance. A statistically significant difference in favour of tenofovir disoproxil 245 mg (as fumarate) was seen in the time-weighted average change from baseline at week 24 (DAVG24) for CD4 count (+13 cells/mm3 for tenofovir disoproxil 245 mg (as fumarate) versus -11 cells/mm3 for placebo, p-value = 0.0008). The antiviral response to tenofovir disoproxil fumarate was durable through 48 weeks (DAVG48 was -0.57 log₁₀ copies/ml, proportion of patients with HIV-1 RNA below 400 or 50 copies/ml was 41% and 18% respectively). Eight (2%) tenofovir disoproxil 245 mg (as fumarate) treated patients developed the K65R mutation within the first 48 weeks.

The 144-week, double-blind, active controlled phase of study GS-99-903 evaluated the efficacy and safety of tenofovir disoproxil 245 mg (as fumarate) versus stavudine when used in combination with lamivudine and efavirenz in HIV-1 infected patients naïve to antiretroviral therapy. The mean baseline CD4 cell count was 279 cells/mm3, the mean baseline plasma HIV-1 RNA was 4.91 log₁₀ copies/ml, 19% of patients had symptomatic HIV-1 infection and 18% had AIDS. Patients were stratified by baseline HIV-1 RNA and CD4 count. Forty-three percent of patients had baseline viral loads> 100,000 copies/ml and 39% had CD4 cell counts < 200 cells/ml.

By intent to treat analysis (Missing data and switch in antiretroviral therapy (ART) considered as failure), the proportion of patients with HIV-1 RNA below 400 copies/ml and 50 copies/ml at 48 weeks of treatment was 80% and 76% respectively in the tenofovir disoproxil 245 mg (as fumarate) arm, compared to 84% and 80 % in the stavudine arm. At 144 weeks, the proportion of patients with HIV-1 RNA below 400 copies/ml and 50 copies/ml was 71% and 68% respectively in the tenofovir disoproxil 245 mg (as fumarate) arm, compared to 64% and 63% in the stavudine arm.

The average change from baseline for HIV-1 RNA and CD4 count at 48 weeks of treatment was similar in both treatment groups (-3.09 and -3.09 log₁₀ copies/ml; +169 and 167 cells/mm3 in the tenofovir disoproxil 245 mg (as fumarate) and stavudine groups, respectively). At 144 weeks of treatment, the average change from baseline remained similar in both treatment groups (-3.07 and -3.03 log₁₀ copies/ml; +263 and +283 cells/mm3 in the tenofovir disoproxil 245 mg (as fumarate) and stavudine groups, respectively). A consistent response to treatment with tenofovir disoproxil 245 mg (as fumarate) was seen regardless of baseline HIV-1 RNA and CD4 count.

The K65R mutation occurred in a slightly higher percentage of patients in the tenofovir disoproxil fumarate group than the active control group (2.7% versus 0.7%). Efavirenz or lamivudine resistance either preceded or was coincident with the development of K65R

in all cases. Eight patients had HIV that expressed K65R in the tenofovir disoproxil 245 mg (as fumarate) arm, 7 of these occurred during the first 48 weeks of treatment and the last one at week 96. No further K65R development was observed up to week 144. From both the genotypic and phenotypic analyses there was no evidence for other pathways of resistance to tenofovir.

Anti-Hepatitis B Virus Activity In Vitro: The in vitro antiviral activity of tenofovir against laboratory strains and clinical isolates of HBV was assessed in HepG2 cells. The EC₅₀ values for tenofovir were in the range 0.06 to 1.5 μM . Tenofovir diphosphate inhibits recombinant HBV polymerase with a Ki (inhibition constant) of 0.18 μM . In *in vitro* drug combination studies of tenofovir with nucleoside anti-HBV reverse transcriptase inhibitors lamivudine, telbivudine and entecavir, additive anti-HBV activity was observed. Additive to slight synergistic effects were observed with the combination of tenofovir and emtricitabine.

5.2 Pharmacokinetic properties

Tenofovir disoproxil fumarate is a water soluble ester prodrug which is rapidly converted *in vivo* to tenofovir and formaldehyde.

Tenofovir is converted intracellularly to tenofovir monophosphate and to the active component, tenofovir diphosphate.

Absorption

Following oral administration of tenofovir disoproxil fumarate to HIV infected patients, tenofovir disoproxil fumarate is rapidly absorbed and converted to tenofovir. Administration of multiple doses of tenofovir disoproxil fumarate with a meal to HIV infected patients resulted in mean (%CV) tenofovir C_{max}, AUC0, and C_{min} values of 326 (36.6%) ng/ml, 3,324 (41.2%) ng-hr/ml and 64.4 (39.4%) ng/ml, respectively. Maximum tenofovir concentrations are observed in serum within one hour of dosing in the fasted state and within two hours when taken with food. The oral bioavailability of tenofovir from tenofovir disoproxil fumarate in fasted patients was approximately 25%. Administration of tenofovir disoproxil fumarate with a high fat meal enhanced the oral bioavailability, with an increase in tenofovir AUC by approximately 40% and C_{max} by approximately 14%. Following the first dose of tenofovir disoproxil fumarate in fed patients, the median C_{max} in serum ranged from 213 to 375 ng/ml. However, administration of tenofovir disoproxil fumarate with a light meal did not have a significant effect on the pharmacokinetics of tenofovir.

Distribution

Following intravenous administration the steady-state volume of distribution of tenofovir was estimated to be approximately 800 ml/kg. After oral administration of tenofovir disoproxil fumarate, tenofovir is distributed to most tissues with the highest concentrations occurring in the kidney, liver and the intestinal contents (preclinical studies). *In vitro* protein binding of tenofovir to plasma or serum protein was less than 0.7 and 7.2%, respectively, over the tenofovir concentration range 0.01 to 25 $\mu\text{g/ml}$.

Biotransformation

In vitro studies have determined that neither tenofovir disoproxil fumarate nor tenofovir are substrates for the CYP450 enzymes. Moreover, at concentrations substantially higher (approximately 300-fold) than those observed *in vivo*, tenofovir did not inhibit *in vitro* drug metabolism mediated by any of the major human CYP450 isoforms involved in drug biotransformation (CYP3A4, CYP2D6, CYP2C9, CYP2E1, or CYP1A1/2). Tenofovir disoproxil fumarate at a concentration of 100 $\mu\text{mol/l}$ had no effect on any of the CYP450 isoforms, except CYP1A1/2, where a small (6%) but statistically significant reduction in metabolism of CYP1A1/2 substrate was observed. Based on these data, it is unlikely that clinically significant interactions involving tenofovir disoproxil fumarate and medicinal products metabolised by CYP450 would occur.

Elimination

Tenofovir is primarily excreted by the kidney by both filtration and an active tubular transport system with approximately 70-80% of the dose excreted unchanged in urine following intravenous administration. Total clearance has been estimated to be approximately 230 ml/h/kg (approximately 300 ml/min). Renal clearance has been estimated to be approximately 160 ml/h/kg (approximately 210 ml/min), which is in excess of the glomerular filtration rate. This indicates that active tubular secretion is an important part of the elimination of tenofovir. Following oral administration the terminal half-life of tenofovir is approximately 12 to 18 hours.

Linearity/non-linearity

The pharmacokinetics of tenofovir were independent of tenofovir disoproxil fumarate dose over the dose range 75 to 600 mg and were not affected by repeated dosing at any dose level.

Age and gender

Limited data on the pharmacokinetics of tenofovir in women indicate no major gender effect.

Pharmacokinetic studies have not been performed in children and adolescents (under 18) or in the elderly (over 65).

Pharmacokinetics has not been specifically studied in different ethnic groups.

Renal impairment

Pharmacokinetic parameters of tenofovir were determined following administration of a single dose of tenofovir disoproxil 245 mg to 40 non-HIV infected patients with varying degrees of renal impairment defined according to baseline creatinine clearance (CrCl) (normal renal function when CrCl> 80 ml/min; mild with CrCl = 50- 79 ml/min; moderate with CrCl = 30-49 ml/min and severe with CrCl = 10-29 ml/min). Compared with patients with normal renal function, the mean (%CV) tenofovir exposure increased from 2,185 (12%) ng-h/ml in subjects with CrCl> 80 ml/min to respectively 3,064 (30%) ng-h/ml, 6,009 (42%) ng-h/ml and 15,985 (45%) ng-h/ml in patients with mild, moderate and severe renal impairment. The dosing recommendations in patients with renal impairment, with increased dosing interval, are expected to result in higher peak plasma concentrations and lower C_{min} levels in patients with renal impairment compared with patients with normal renal function.

The clinical implications of this are unknown.

In patients with end-stage renal disease (ESRD) (CrCl < 10 ml/min) requiring haemodialysis, between dialysis tenofovir concentrations substantially increased over 48 hours achieving a mean C_{max} of 1,032 ng/ml and a mean AUC_(0-48h) of 42,857 ng-h/ml.

It is recommended that the dosing interval for tenofovir disoproxil 245 mg (as fumarate) is modified in patients with creatinine clearance < 50 ml/min or in patients who already have ESRD and require dialysis (see 4.2).

The pharmacokinetics of tenofovir in non-haemodialysis patients with creatinine clearance < 10 ml/min and in patients with ESRD managed by peritoneal or other forms of dialysis

have not been studied.

Hepatic Impairment

A single 245 mg dose of tenofovir disoproxil was administered to non-HIV infected patients with varying degrees of hepatic impairment defined according to Child-Pugh-Turcotte (CPT) classification. Tenofovir pharmacokinetics were not substantially altered in subjects with hepatic impairment suggesting that no dose adjustment is required in these subjects. The mean (%CV) tenofovir C_{max} and AUC0_∞ values were 223 (34.8%) ng/ml and 2,050 (50.8%) ng-hr/ml, respectively, in normal subjects compared with 289 (46.0%) ng/ml and 2,310 (43.5%) ng-hr/ml in subjects with moderate hepatic impairment, and 305 (24.8%) ng/ml and 2,740 (44.0%) ng-hr/ml in subjects with severe hepatic impairment.

Intracellular pharmacokinetics

In non-proliferating human peripheral blood mononuclear cells (PBMCs) the half-life of tenofovir diphosphate was found to be approximately 50 hours, whereas the half-life in phytohaemagglutinin-stimulated PBMCs was found to be approximately 10 hours.

6. PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Core tablet: croscarmellose sodium, lactose monohydrate, magnesium stearate and microcrystalline cellulose.

Film coat: FD & C Blue #2/Indigo carmine aluminium lake, hypromellose, lactose monohydrate, titanium dioxide and triacetin/glycerol triacetate

6.2 Nature and contents of container

HDPE bottle of 30'S.

6.3 Special precautions for storage and disposal

Do not store above 30°C. Store in the original container. Protect from light.

No special requirements.

Any unused product or waste material should be disposed off in accordance with local requirements.

Safe disposal instructions about the desiccant: dessicant bag or its contents must not chewed, swallowed or torn. It should be disposed off intact.

7. Shelf Life

36 months



Mfd. by:

Mylan Laboratories Limited
F-4 & F-12, MIDC, Malegaon, Sinnar,
Nashik - 422 113, Maharashtra, INDIA.

Product Registration Holder (PRH)
Mylan Healthcare Sdn. Bhd. (1161821-X)
15-03 & 15-04, Level 15, Imazium,
No. 8, Jalan SS 21/37, Damansara Uptown,
47400, Petaling Jaya, Selangor, Malaysia.

March 2024

Mylan Laboratories Limited Packaging Development	Artwork Implementation Schedule Check (✓) whichever is applicable (Approval is not valid without following details)					Date of Issue										
						Date of Return		Issued By								
						Material Code		75091542	Supersedes	75065232	Market	Malaysia				
	[] New Component [] Immediately (Stock of superseded component to be destroyed, if applicable.) [] After consumption of existing (superseded) stock. [] Other (Specify)					Description		LIT. RICOVIR TABS 300 mg (36 Months) MY V2								
						Component		Printed Literature	Actual Size	Flat- 272 x 312 mm; Folded- 34 x 39 mm						
					Substrate		60gsm Century Maplitho Paper									
					Design & Style		Supply in Folded form as Proposed Size (with tape)									
					Reason for Issue		Change in Mylan Healthcare address									
<i>Sign.</i> dd/mm/yy <i>Job Function</i>					<i>Sign.</i> dd/mm/yy <i>Job Function</i>	<i>Sign.</i> dd/mm/yy <i>Job Function</i>	Printing Pantone Nos	1	BLACK	2	NA	3	NA	4	NA	
					Non Printing			Die Line		NA		NA		NA		NA
Date		01/06/2022	dd/mm/yy	dd/mm/yy	dd/mm/yy	dd/mm/yy	Prepared By		Checked By			Approved By				
Remarks		Revised A/w	x	x	x	x	Packaging Development	Packaging Development	Production		Regulatory Affairs		Quality Assurance			
SOP-000565164-FORM-000565208-A01-03-01-20			Final Date		dd/mm/yy											